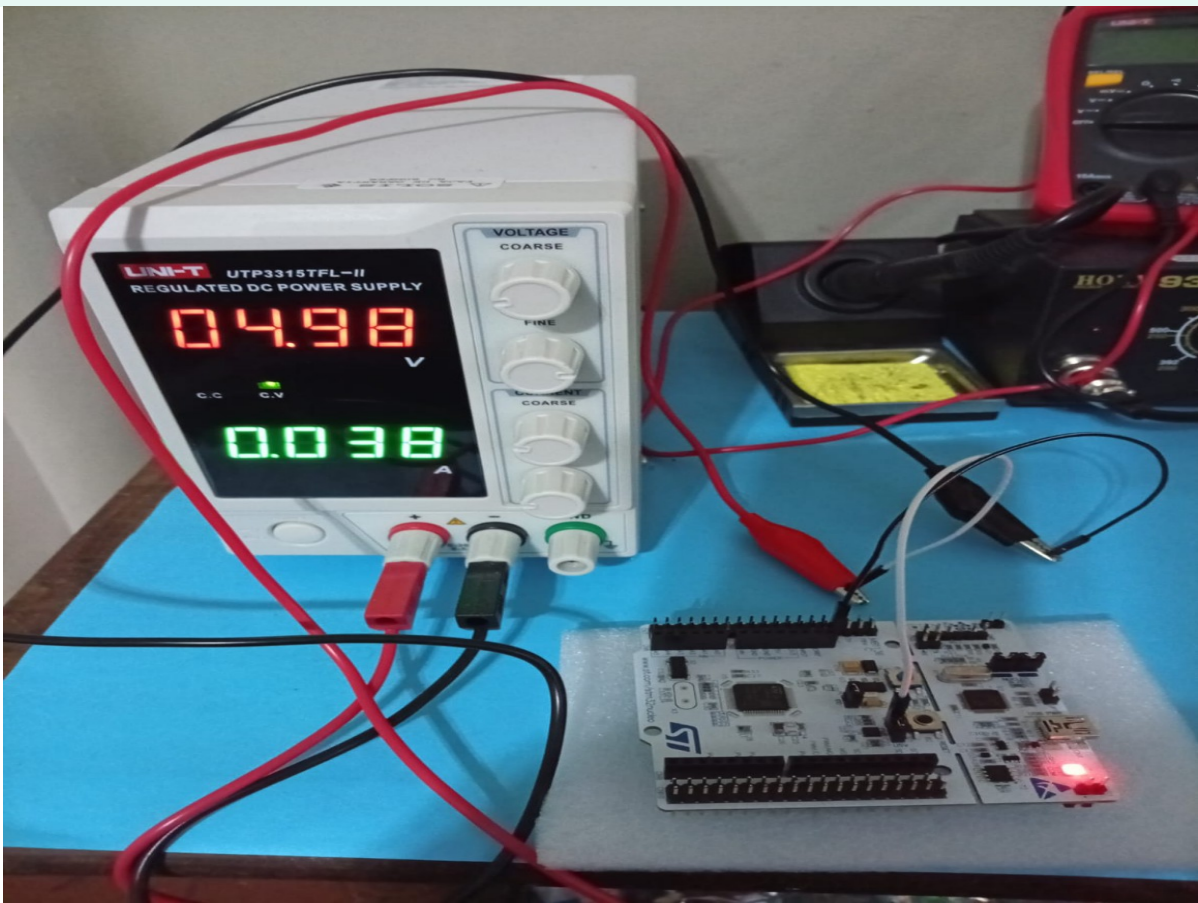


OSH

Physical Cross-Architecture Evidence

A public whitepaper on portability, physical execution and measured embedded evidence

Public disclosure boundary. This paper presents observable behavior, hardware targets, test methodology, measured results and artifact identities. It does not disclose the proprietary OSH kernel, internal algorithms, source architecture, decision equations, semantic mechanisms, training methods, or other black-box implementation details.



STM32F446RE physical ARM validation under external regulated power.

Organization	OSH Global Systems, Inc.
Edition	October 2026
Scope	PC → portable substrate → ESP32-S3 physical → STM32F446RE physical ARM
Public claim	Measured cross-architecture embedded evidence
Excluded	Proprietary kernel and black-box implementation details

Abstract

OSH is an embedded architecture being developed for deterministic, local operation without dependence on cloud inference, external APIs, or an LLM runtime. The engineering question addressed by this whitepaper is not whether a codebase can merely be compiled for different processors, but whether expected behavior can be preserved as execution moves across materially different computational substrates and into physical embedded hardware.

The evidence chain summarized here progresses from PC reference work and a Universal Substrate Benchmark to physical ESP32-S3 execution and, most recently, physical STM32F446RE ARM Cortex-M4 execution. The ARM milestone adds native execution, a 30-case parity gate, a 100,000-cycle physical DAC-to-ADC closed loop, external oscilloscope observation, artifact hashing, and board-level input-power observation.

All claims are intentionally bounded to what was measured. Compilation is not treated as execution. Board-level power is not presented as isolated OSH-core power. Oscilloscope cursor coincidence is not described as zero latency.

Stage	Evidence	Status
PC	Reference execution and benchmark baseline	Demonstrated
Universal Substrate Benchmark	Frozen Tier-A substrate across multiple ISA profiles; RV32IMF golden-case reproduction	Demonstrated
ESP32-S3	Physical embedded reproduction and HAL validation	Demonstrated
STM32F446RE	Physical ARM execution, parity, 100K loop, scope and power evidence	Demonstrated
Physical RISC-V	Next cross-architecture hardware milestone	Next

1. Why physical portability is a separate engineering problem

Portability has distinct levels. Source compatibility demonstrates that a toolchain can accept code. Object generation demonstrates that target instructions can be emitted. Emulator execution demonstrates behavior under a model. Physical execution adds real instruction-set behavior, floating-point behavior, memory organization, clocks, peripherals, electrical interfaces and target-runtime constraints.

OSH therefore treats 'compiled for another architecture' and 'verified on another physical architecture' as different claims.

Level	Demonstrates	Does not by itself demonstrate
Compile	Target toolchain accepts source	Correct runtime behavior
Link / image	Loadable target artifact exists	Physical execution
Emulated execution	Behavior under modeled target	Electrical/peripheral behavior
Physical execution	Native runtime on real silicon	Universal correctness
Physical parity + instrumentation	Observed behavior plus real I/O evidence	Independent certification

2. ARM physical validation target

The ARM validation used an ST NUCLEO-F446RE board carrying an STM32F446xx MCU with an ARM Cortex-M4 CPU. The target was detected through ST-LINK, programmed through SWD, and the programmed image was verified before reset and execution.

Property	Observed / configured
Board	NUCLEO-F446RE
MCU	STM32F446xx
CPU	ARM Cortex-M4
NVM reported by programmer	512 KB
Programming interface	ST-LINK / SWD
Firmware verification	Download verified successfully
100K image size	67.89 KB

The public evidence concerns target behavior and external interfaces. The proprietary OSH kernel remains outside this disclosure.

3. Structural parity before the physical loop

Before entering the long physical run, the ARM target executed a 30-case structural parity gate. The complete gate passed.

Validation	Result
Parity cases	30 / 30
State parity	PASS
Settle / vortex parity	PASS
HAL parity	PASS
Selected-edge identity parity	PASS
Meaning parity	PASS
Combined parity gate	PASS

4. Physical electrical loop and 100,000-cycle run

The physical ARM experiment connected the STM32 DAC output at A2/PA4 to the ADC input at A0/PA0. The target first tested three electrical levels through this physical path and then executed a 100,000-cycle closed loop.

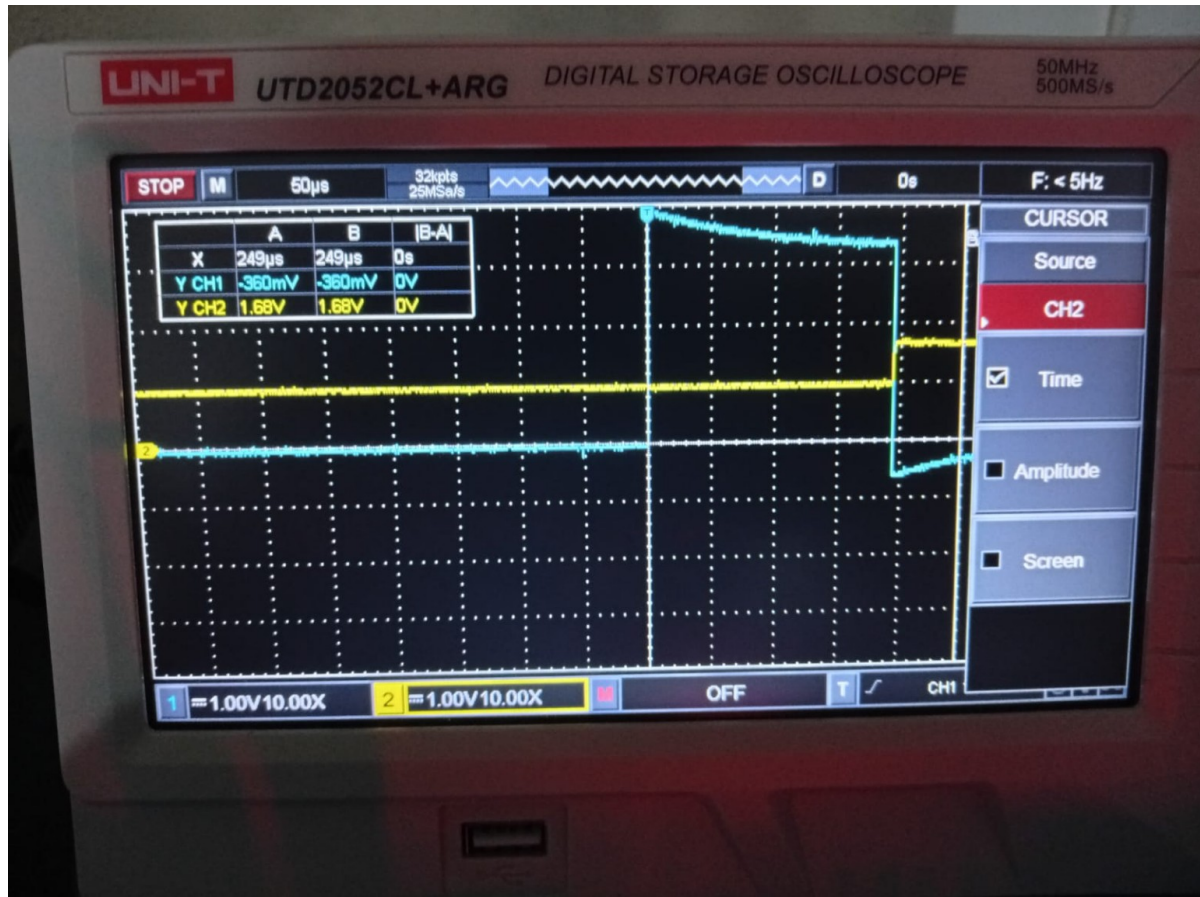
Measurement	Result
Low / mid / high ADC probe	512 / 2053 / 3590
Electrical loop gate	PASS
Cycles	100,000
ADC failures	0
State failures	0
Actions	INC 36 / HOLD 99,934 / DEC 30
ADC range	1243 – 2759
DAC range	1280 – 2720
Max DAC↔ADC mismatch	43 counts
Final ADC / DAC / reference	2343 / 2336 / 2662
Final error	319 counts
Elapsed time	24.937 s
Physical 100K gate	PASS
Overall ARM 100K	PASS

A real analog output left the MCU, crossed an external electrical connection and returned through the ADC. This is not presented as a simulation-only loop.

5. Runtime timing and external instrumentation

Timing metric	Result
Minimum	209.500 μ s
Maximum	241.875 μ s
Mean	241.250 μ s
Cycle-counter values	3352 / 3870 / 3860 cycles (min / max / mean)

A separate external oscilloscope experiment observed an instrumented execution window independently of the UART report.

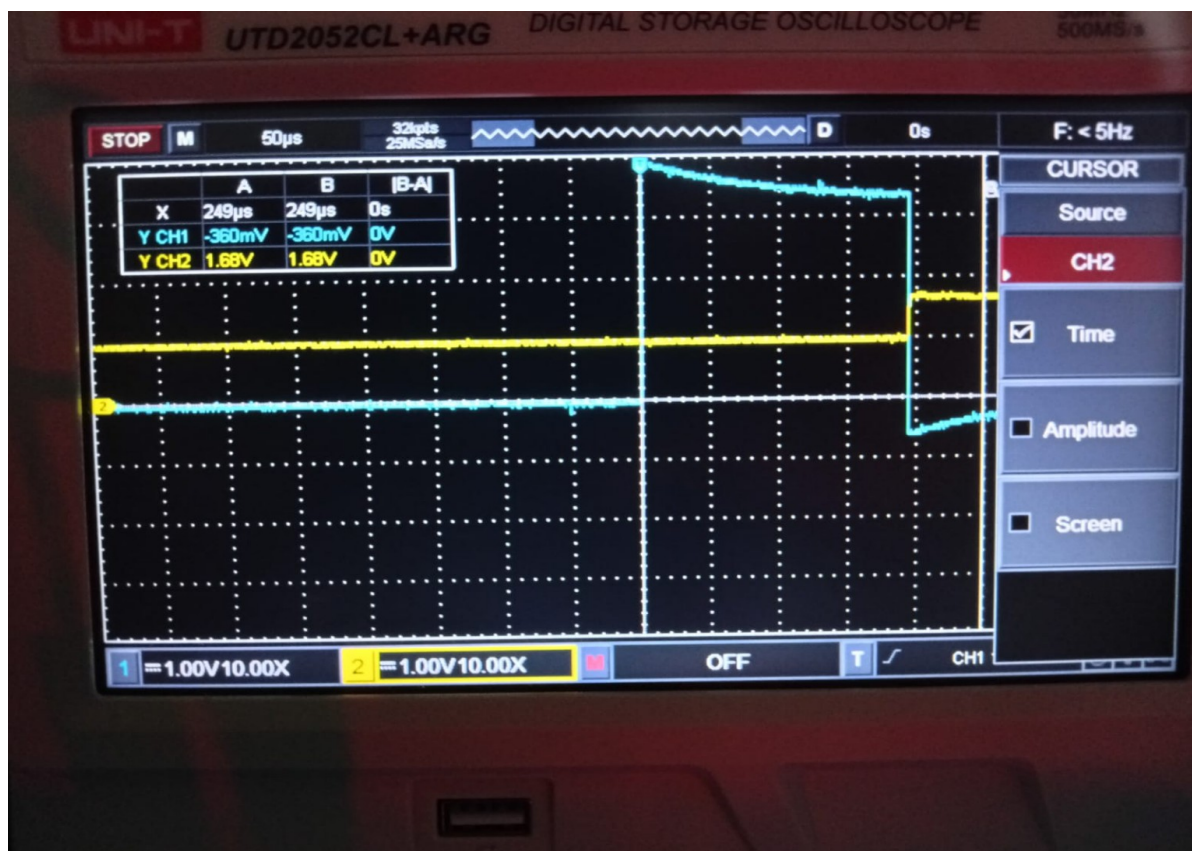


External oscilloscope capture: CH1 marks the instrumented execution window; CH2 observes the analog-output event.

The externally measured CH1 pulse width was **226 μ s** at 50 μ s/div. Because the scope firmware and the 100K timing window are not asserted to be identical instrumentation boundaries, the result is presented as an external measurement consistent with the same timing scale, not as a calibration identity.

6. External causal-path observation

A dual-channel oscilloscope experiment observed the execution marker and analog output simultaneously. For visibility in this instrumentation-only experiment, the DAC step was deliberately enlarged; its amplitude is therefore not the normal 100K control step.

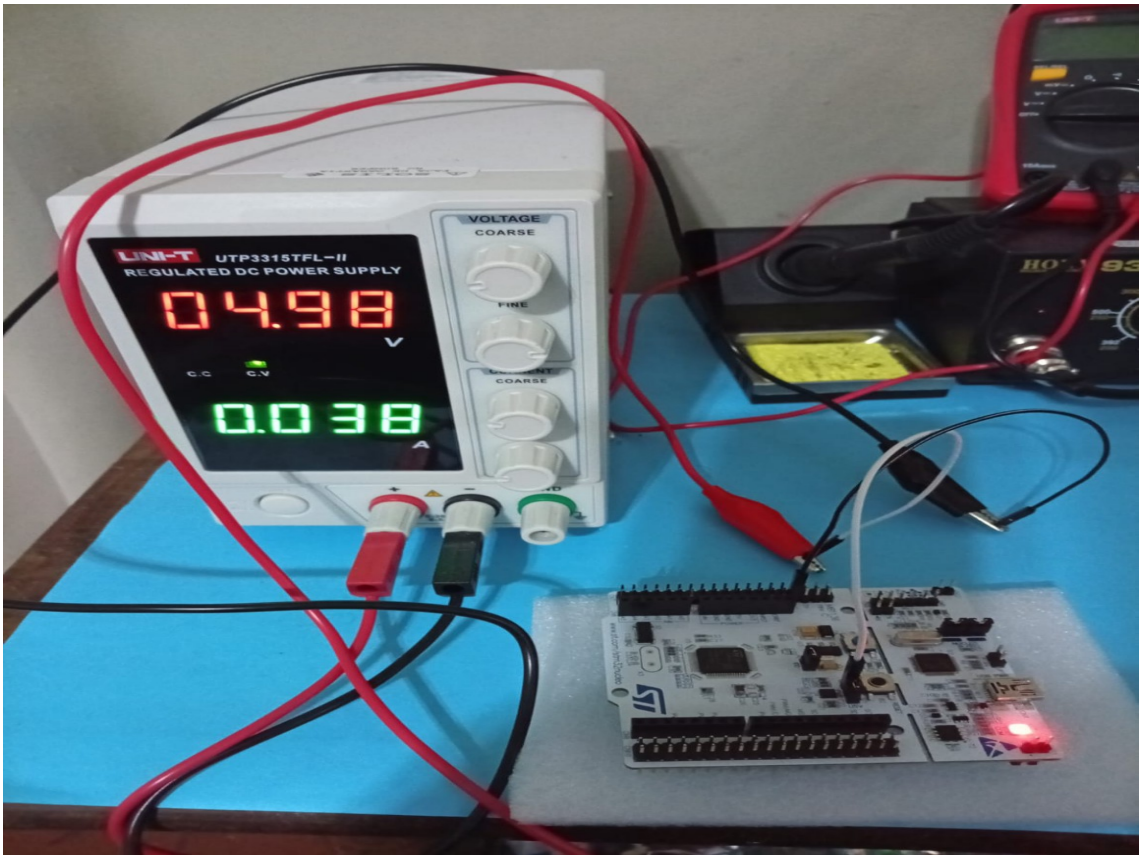


Dual-channel frozen trace used to observe the temporal relationship between execution marker and analog-output transition.

The final cursor capture showed $A = 249 \mu\text{s}$, $B = 249 \mu\text{s}$ and $|B-A| = 0 \text{ s}$. The correct interpretation is coincident within the time resolution of the captured trace. This is not a claim of zero physical latency.

7. Board-level input power observation

After the 100K firmware was restored and verified, USB power was removed and the NUCLEO was powered from an external regulated UNI-T supply at 4.98 V. During the run window, approximately 27–38 mA was observed at the board input.



External power observation. The photographed instant shows 4.98 V and 0.038 A.

Observed board current	Calculated input power at 4.98 V
27 mA	134.5 mW
38 mA	189.2 mW

A video of approximately 28.2 seconds covered essentially the full ~24.94-second 100K execution window. The reported range is a board-level input-power observation during the ARM validation run. It includes the NUCLEO platform and associated electronics and is not isolated OSH-core power or an energy-efficiency certification.

8. Artifact integrity

Key ARM validation artifacts were frozen and identified with SHA-256 hashes. Hash publication supports later byte-identity checks without disclosing the proprietary kernel.

Artifact	SHA-256
ARM validation ELF	9cd4c56ab5c29bb150900c96abff3328fdeb1b28e13b639fd97fbb0f6a7a0645
ARM validation BIN	336788116049fcfcbee03210c48b32cab92c5670216e59765280c016440fcaae
ARM runner	906712ae003c352728134937182e30850befea3bc06ef661e21181022048f6fa
UART bridge	4759deeee57d592bbe5cd234e8a6bf46206b649652b08c26d3021387b7417552
UART PASS record	c58ec37b2995e00e1f71f66659d5c1122d613fe1d959780c9524d5e5a4b63720
Board-ID record	54e38fa69c77f8e66b24215d76b48a1228f4112204c88b3a9e62c49a37d27a75

9. Public evidence boundary

Public evidence	Remains proprietary
Target hardware and toolchain identity	OSH kernel implementation
Test topology and external electrical path	Internal algorithms and equations
Measured parity and gate results	Semantic-state construction mechanisms
Cycle counts, timing and failure counts	Decision internals and proprietary selectors
Oscilloscope observations	Source-level black-box architecture
Board-level input-power observation	Private adaptation mechanisms
Artifact hashes	Confidential production code

The boundary is intentional: externally testable claims are made visible while the mechanism that produces them remains proprietary.

10. Conclusion and next milestone

The ARM result changes the status of OSH portability evidence. ARM is no longer represented only as a compilation target: the STM32F446RE physically executed the validation path, passed the frozen 30-case parity gate, operated a real DAC-to-ADC electrical loop for 100,000 cycles with zero reported ADC or state failures, and produced externally observable timing and output events.

This does not establish universal hardware portability. It establishes an additional physical architecture beyond the prior embedded evidence and strengthens the progression from PC and portable-substrate verification toward multi-architecture physical deployment.

Next milestone

The next high-value evidence point is physical RISC-V execution using the same discipline: frozen validation boundary, target identification, native build, physical execution, parity/golden-case comparison, long-run stability, external instrumentation and artifact hashes.

Public technical document. Proprietary implementation details intentionally omitted.